

A Wideband Sub-6-GHz Transceiver Front End for 5G Base Stations in 22-nm FD-SOI CMOS

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Abstract—A fully integrated, non-frequency-translating, low-impedance transceiver (TRX) front end for cellular base stations (BSs) covering 1.25–5.5 GHz is presented. The transmitter (TX) features two variable gain feedback pre-power amplifier (PPA) stages, a power amplifier (PA), and an RF notch filter based on frequency translation for suppressing spurious signals. The receiver (RX) features a low noise amplifier (LNA), a variable gain amplifier (VGA), and an equalizer supporting three concurrent frequency bands with orthogonal frequency and gain tuning. The PA and LNA use a 12.5- Ω differential impedance, primarily to boost PA output power and bandwidth (BW). Special emphasis is put on the thermal analysis of the PA and the techniques to improve the cooling. When operating with full PA supply (7.4 V), including antenna-switch and package losses, the TX reaches a record saturated output power (P_{sat}) of 38.1 dBm at 3.4 GHz, and for a 100-MHz 16-QAM orthogonal frequency-division multiplexing (OFDM) signal at 2.7 GHz, the TX delivers an output power of 32.6 dBm with an error vector magnitude (EVM) of –17.6 dB, with a complete TX efficiency of 13.5%. With a reduced PA supply of 3.8 V, an 8 $\times$ 100 MHz 256-QAM OFDM signal with 21.6 dBm was measured with EVM < –31.1 dB. The TX filter achieves about 40-dB notch depth at 1–6-GHz notch frequencies, and the 20- and 30-dB rejection BW can be controlled from 1 to 10 MHz and from 0.4 to 4 MHz, respectively. The front end has an RX noise figure (NF) of 1.3–2 dB, while the third-order intercept point (IIP3) is above –12.4 dBm, at maximum gain (29 dB) over the frequency range of 1.55–4.8 GHz. The passive equalizer improves the RX interference handling and is analyzed using closed-form expressions. The active part of the die, manufactured in 22-nm fully depleted silicon on insulator (FD-SOI) CMOS technology, occupies 6.6 mm².

Index Terms—Equalizer, front end, high power, linear, low impedance, low noise amplifier (LNA), power amplifier (PA), receiver (RX), thermal, transceiver (TRX), wideband.

I. INTRODUCTION

THERE is a recent trend in sub-6-GHz cellular base station (BS) equipment to use antenna array systems (AASs) with an increasing number of antenna elements. This direction is pointed out by 5G millimeter-wave (mm-wave) BS that must overcome severe path losses, using many antenna elements

to increase the effective isotropic radiated power (EIRP) and effective isotropic sensitivity [1]. For a sub-6-GHz BS though, the benefits on the transmitter (TX) side of increased AAS size are foremost reduced transmit power per transceiver (TRX) output so that technologies such as CMOS can be used for power amplifiers (PAs) to support a high integration level and increased array gain so that the total radiated power (TRP) can be lowered without reducing the EIRP, providing significant power saving in the BS. For a fixed PA efficiency the total PA power consumption is expected to be inversely proportional to the AAS size, thereby enabling a more sustainable cellular network [2].

Nevertheless, due to longer wavelengths, a sub-6-GHz AAS will have fewer antenna elements than one for mm-waves, still resulting in a stringent output power requirement per PA, which makes integration in CMOS a challenge. The main issue is the poor ability of CMOS transistors to withstand high voltage stress. The PA must therefore drive a low impedance so that high power can be delivered without high voltage. Another benefit of using low load impedance is that it effectively increases the bandwidth (BW) in the presence of shunt reactance, which is highly desirable since a TRX circuit can then support wideband operation covering multiple frequency bands, which reduces the number of product variants, eases deployment, and enables support of multi-band transmissions. A downside of having low TRX impedance is difficulty in reaching low receiver (RX) noise figure (NF), as lower impedance reduces the input signal voltage, which increases the low noise amplifier (LNA) power consumption. That, however, is less problematic as the power consumption for RX is typically significantly lower than for TX, and in a BS, it becomes further reduced due to the uplink (UL) typically having a much lower duty cycle than the down-link (DL). Furthermore, to handle challenging blocking scenarios, high RX linearity is important; lower on-chip voltage signal levels due to the lower impedance are then highly beneficial, and they also make the LNA less vulnerable to damage from strong signals transmitted to the antenna.

Radio frequency digital-to-analog converters (RF-DACs) and analog-to-digital converters (RF-ADCs) can convert directly between digital signals and modulated RF signals, which enables TRXs with excellent reconfigurability and wideband/multiband capabilities, and recent advances have made them a viable option for the sub-6-GHz frequency range [3], [4]. To fully utilize the flexibility offered by RF-DACs and RF-ADCs, they should be matched with wide-

Received 21 November 2024; revised 4 February 2025; accepted 10 March 2025. Date of publication 1 April 2025; date of current version 1 July 2025. This work was supported by Ericsson. This article was approved by Associate Editor Yang Zhang. (Christian Elgaard and Mohammed Abdulaziz contributed equally to this work.) (Corresponding author: Christian Elgaard.)

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Color versions of one or more figures in this article are available at <https://doi.org/10.1109/JSSC.2025.3552278>.

Digital Object Identifier 10.1109/JSSC.2025.3552278

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band TRX front ends. Filtering then poses problems in both RX and TX, as filtering at RF is much more challenging than at baseband, especially for wideband and multi-band RF signal interfaces, where implementation of RF bandpass filters with multiple tunable passbands is required. To reduce power requirements of the RF-ADC, however, RX filtering with on-chip multi-band RF filtering and gain equalization is needed, and TX filtering is important to reduce the spur level.

In recent years, multiple CMOS TRXs for sub-6-GHz communication have been published [5], [6], [7], [8], [9]. Unfortunately, all lack some key front-end parts to make them fully integrated, such as having external PAs, no integrated transmit/receive switch (TRX-Switch), and in some cases also requiring external LNAs. Furthermore, they all target user equipment (UE) implementation and have multiple parallel TX/RX paths to support multiple bands. In BS implementations, the RF front end typically consists of multiple integrated circuits and discrete modules, using different semiconductor technologies such as gallium nitride (GaN), gallium arsenide (GaAs), and silicon for different parts. Having multiple chips and discrete modules increases the cost of material and assembling and jeopardizes the yield.

This work presents a highly integrated TRX implemented in 22-nm FDSOI CMOS aimed for cellular sub-6-GHz BS. The TRX consists of a low-impedance high-power linear wideband TX [10] equipped with a frequency-translated TX filter, and a low-impedance low-NF wideband RX equipped with a tunable triple-band RF equalizer [11]. Important contributions are the design and analysis of an integrated high-power PA in 22-nm CMOS technology using laterally-diffused metal-oxide semiconductor (LDMOS) transistors, including thermal considerations, a frequency-translated notch filter with programmable cancellation and BW, a wideband multi-band filter and equalizer, an LNA with low input impedance and wide BW, a high dynamic range and wideband variable gain amplifier (VGA), and a highly linear wideband pre-PA (PPA) using current amplifier feedback.

II. SYSTEM CONTEXT

Table I compares the UL/DL link budget at 3 GHz for three different building practice options for a 5G BS TRX front end. The first option is a highly integrated TRX system with 64 antenna elements each connected to a TRX channel, the second option uses 16 antenna elements each connected to an RX input and fed by an external PA, and the third option uses 64 antenna elements arranged in groups (sub-arrays) of 4, where each sub-array is connected to an RX input and an external PA. All three options are assumed to have the same TX EIRP of 68 dBm, the same TX efficiency of 10%, and an RX NF of 4 dB, but with different PA output power. As can be seen in Table I, the choice of option will highly affect the TX power consumption and also the UL performance. More antenna elements are beneficial as it increases the array gain and the TRP can then be reduced, which lowers BS power consumption, and as can be seen, the option with only 16 antennas has the by far highest TX power consumption. Increased sub-array sizes can be used to increase the number

TABLE I
BS BUILDING PRACTICE COMPARISON, 68-dBm EIRP
AT 3 GHz, LINK BUDGET PER POLARIZATION

Building Practice	Int. TRX		Int. TRX + Ext. PA		Int. TRX + Ext. PA	
Frequency [GHz]	3		3		3	
Free Space Dist. [km]	10		10		10	
Antennas BS, UE	8 x 8, 1		4 x 4, 1		8 x 8, 1	
Antennas per PA/LNA	1		1		4	
Antenna Size [m ²]	0.16		0.04		0.16	
Link	DL	UL	DL	UL	DL	UL
BW [MHz]	100	16	100	4	100	10
QAM-Mod	64	4	64	4	64	4
Raw Data Rate [Mbit/s]	600	32	600	8	600	20
TX Element Pout [dBm]	30	27	42	27	32	27
TX Routing Loss [dB]	1	1	1	1	3	1
TRP [dBm]	47	26	53	26	47	26
TX Antenna Gain [dB]	3	3	3	3	3	3
EIRP [dBm]	68	29	68	29	68	29
Path Loss [dB]	122	122	122	122	122	122
RX Antenna Gain [dB]	3	3	3	3	3	3
RX Routing Loss [dB]	1	1	1	1	1	3
RX B.F. Gain [dB]	0	18	0	12	0	18
Received Power [dBm]	-52	-73	-52	-79	-52	-75
Required SNR [dB]	27	15	27	15	27	15
Margin [dB]	10	10	10	10	10	10
RX noise [dBm]	-89	-98	-89	-104	-89	-100
RX NF [dB]	5	4	5	4	5	4
BW Noise Increase [dB]	80	72	80	66	80	70
kT [dBm/Hz]	-174	-174	-174	-174	-174	-174
TX DC Cons.⁽¹⁾ [W]	640	5	2536	5	1014	5

Int. TRX = Integrated Transceiver, Ext. PA = External PA, B.F. = Beamforming Gain, ⁽¹⁾ Assuming 10% TX efficiency for all building practices.

- TRP = TX Element Pout + 10Log₁₀(NbrofPAs) - TX Routing Loss
- EIRP = TRP + 10Log₁₀(NbrofTXAntennas) + TX Antenna Gain
- RX B.F. Gain = 10Log₁₀(NbrofRXAntennas)
- Received Power = EIRP - Path Loss + RX Antenna Gain - RX Routing Loss + RX B.F. Gain
- Required SNR = 15 dB for 4-QAM and 27 dB for 64-QAM
- Margin = Received Power - Required SNR - RX noise
- RX Noise = kT + BW Noise Increase + RX NF
- BW Noise Increase = 10Log₁₀(BW)

of antenna elements while still using a limited number of external PAs, but it increases the losses between the antennas and the TRX. This is reflected in Table I by an increased PA output power requirement to reach the EIRP of 68 dBm and an unavoidable reduced UL SNR. As can be seen in Table I, the option of a highly integrated TRX is most attractive in terms of TX power consumption and UL data rate. Moreover, larger sub-arrays also reduce spatial selectivity due to increased grating lobes. Another strong motivation for larger antenna arrays with integrated TRXs is the reduced cost by avoiding external PAs. The goal of this work is therefore to enable such a building practice using TRXs in integration-friendly CMOS technology, with integrated PAs supporting the required power levels with modulated signals.

With continued network deployment and densification, the requirements on RX large-signal handling will likely increase, calling for RXs for network equipment with low input impedance to reduce the input signal voltage levels. A particularly challenging scenario for the RX is when a BS should receive an over-the-air synchronization (OAS) signal, for improved synchronization accuracy, during the DL time slot [12]. A neighboring BS may then still be transmitted in an adjacent channel, causing the RX to receive a large signal.

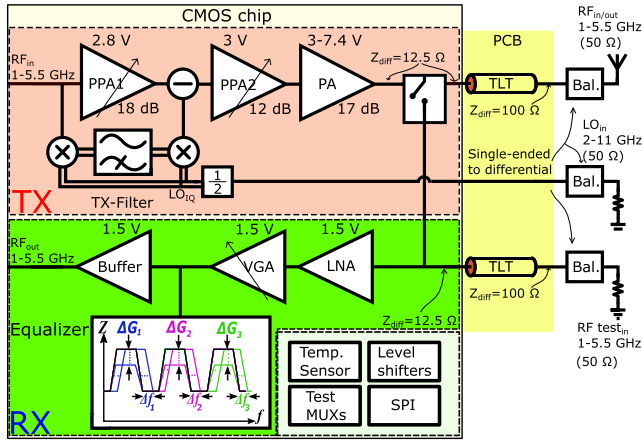


Fig. 1. TRX architecture with nominal gain values (at 3 GHz).

A neighboring co-sited (on the same mast) BS with a TRP of 53 dBm and an isolation of 45 dB would result in 8 dBm at the LNA input. Assuming a peak-to-average (PAR) of 9 dB, the LNA must then handle 17-dBm peak input power without breaking. In a 50- Ω interface, 17 dBm corresponds to 2.2-V peak voltage, while a 6.25- Ω interface results in 0.8 V. In a differential signal interface with 6.25- Ω impedance per side, the peak voltage becomes 0.56 V. In advanced CMOS nodes, thin-oxide devices used for low NF have a maximum supply voltage below 1 V, with similar gate voltage levels specified to avoid damage. The lower impedance of 6.25 Ω and the differential RX signal reduce the input signal voltage of the LNA transistors by a factor of 4, thus preventing damage in the above scenario.

III. TRX FRONT-END ARCHITECTURE

The TRX front-end architecture is shown in Fig. 1. It is non-frequency translating and the TX part consists of three cascaded amplifiers, two PPAs (PPA1 and PPA2) and the PA, whereas the RX part consists of LNA, VGA, passive multi-band equalizer, and an output buffer. The input to the TX and the output from the RX are 100- Ω differential signals. The TRX RF port has a 12.5- Ω differential impedance, to which the TX output is connected directly and the RX is through a TRX-Switch. The RX also has a test port connected directly to its input to enable bypass of the TRX-Switch in measurements. A three-stage transmission line transformer (TLT), similar to [13], is integrated into the printed circuit board (PCB), transforming the differential 12.5- Ω impedance of the TRX RF port (and the test port) to 100 Ω . An external balun is then used to transform the differential signal to a single-ended 50- Ω signal suitable to interface with test equipment.

The 12.5- Ω differential impedance of the interface to the integrated circuit was chosen to allow a peak power of >38 dBm to be delivered by the PA, at the supply voltage of 7.4 V, without using any on-chip impedance transformation. By performing all impedance transformation off-chip, a wide BW and low loss can be obtained for the TX signal. The impedance transformation is performed by a low-loss TLT, converting 12.5–100- Ω differential impedance over a 1.25–5.5-GHz frequency range. To minimize high-frequency

signal routing in package and PCB, and on-chip-to-off-chip transitions to the integrated TRX-Switch, the RX also has a 12.5- Ω differential impedance. This low impedance results in less input signal voltage, which requires using larger LNA input transistors with more bias current to reduce their input-referred noise voltage, so a low NF can still be achieved. The increased transconductance (g_m) also increases the LNA voltage gain to compensate for the reduced input signal amplitude. A drawback of the lower impedance is that both TX and RX become more vulnerable to parasitic series resistance, which, however, fundamentally becomes low for low-impedance transmission lines, but on the other hand, they become less sensitive to shunt capacitance. A benefit of the low impedance is that interfering signals have lower voltage at the LNA input, reducing the risk for LNA damage due to strong nearby TXs.

The TX is assumed to be preceded by an RF-DAC, which provides an RF input signal, similar to [3]. Spurs in the DL can then originate from the DAC sampling clock. The frequency of the spur to filter, however, is not necessarily equal to the sampling clock frequency but may be an intermodulation product of the clock and the signals. Measurements of the TX spectrum emissions to ensure compliance with the 5G standard are thus used to select the frequency to filter, and to suppress potential out-of-band spurious signals present in the TX input signal, a TX notch filter was designed as a frequency-translated low-pass filter (LPF) path in parallel with PPA1. The TX filter taps off a portion of the input signal, which it down-converts to a quadrature baseband in-phase and quadrature (IQ) signal, then scales and rotates, filters, and up-converts it back to RF, and injects in anti-phase at the output of the PPA1. The filter is narrowband (high Q) at RF, enabling suppression of a small selected portion of the RF signal spectrum. Both PPA1 and PPA2 are VGAs implemented with negative feedback for high linearity. The PA is a differential common source (CS) common gate (CG) stage biased in class AB for high linearity.

In the RX, an LNA with 12.5- Ω input matching first amplifies the input signal, and the VGA then further amplifies it with programmable gain to adapt the signal level to the dynamic range of the ADC. At the VGA output, an equalizer is used to independently adapt the amplitude of three different frequency bands to account for signal fading or asymmetric UL path losses and provide suppression outside those, where blockers may exist. Even a low amount of suppression is valuable in improving the RX linearity and reducing ADC power consumption, where relaxing the dynamic range by 6 dB reduces the ADC power consumption to one fourth, given that the ADC is thermal noise limited, which is typical for high-performing ADCs [14]. As indicated in Fig. 1, the equalizer can change the frequency and gain of, e.g., the first band by a factor of Δf_1 and ΔG_1 . Finally, the output buffer amplifies the signal, providing it to the 100- Ω differential test equipment.

IV. CIRCUIT DESIGN

A. LNA

The key objectives in the LNA design are to achieve low NF and wide BW while being matched to a low impedance.

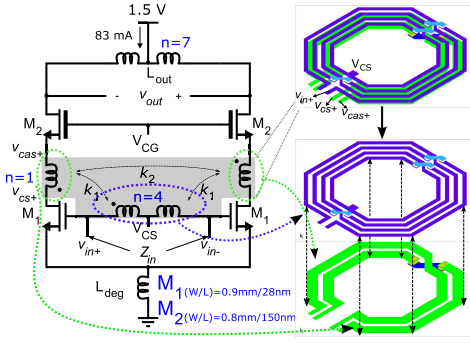


Fig. 2. Schematic of the differential LNA.

High linearity and limited power consumption and chip area are other important requirements. In prior art, transformer feedback has been successfully used to improve the BW and achieve low NF [15], [16], and shunt-to-shunt feedback was used in [16] to improve LNA BW. Low NF requires high g_m of the LNA input transistor, especially when matching to low impedance. In [15], matching to $50\ \Omega$ could be achieved by either decreasing g_m , degrading the available NF, or by increasing the number of the turns in the secondary transformer windings, resulting in increased chip area due to the two large transformers required in the pseudo-differential LNA.

A fully differential LNA was chosen to be compatible with the signals of the differential TX-part and for the usual benefits of differential circuits providing even-order non-linearity cancellation, common-mode interference rejection, and so on. The schematic is shown in Fig. 2, using series-shunt feedback with a 1:4 differential transformer structure with two intertwined single-turn primary windings [15]. A CS-CG stage with the signal current sensed by transformer primaries is inserted in series between the two stages. A common-mode degeneration inductor L_{deg} is used to reduce the common-mode gain, and the LNA is loaded by a seven-turn inductor L_{out} , both inductors resonating with the capacitances at the respective nodes. The input stage uses thin-oxide devices M_1 for low NF and wide BW. The CG stage, on the other hand, uses thick oxide devices M_2 for increased voltage handling, enabling a supply of 1.5 V for improved linearity.

To achieve matching to the $12.5\text{-}\Omega$ differential impedance, the transformer turns ratio n and coupling factor k_1 , and the input transistor g_{m1} , are selected based on the following equation:

$$R_i = \frac{n}{k_1 \cdot g_{m1}} \quad (1)$$

where R_i is the inverse of the single-ended LNA input conductance, close to the real part of half Z_{in} (see Fig. 2). For $n = 4$, $k_1 \approx 1$, and $g_{m1} \approx 0.64\text{ S}$, the single-ended R_i becomes $\approx 6.25\ \Omega$. The high value of g_{m1} is needed to achieve low NF when being matched to such low impedance. Furthermore, this topology is attractive since the NF degradation due to the loss of the transformer winding at the LNA input is minimal. The simulated Q -factor of the inductor is larger than 10, resulting in an equivalent parallel resistance above $420\ \Omega$ across the entire frequency range. Thanks to the low $12.5\text{-}\Omega$

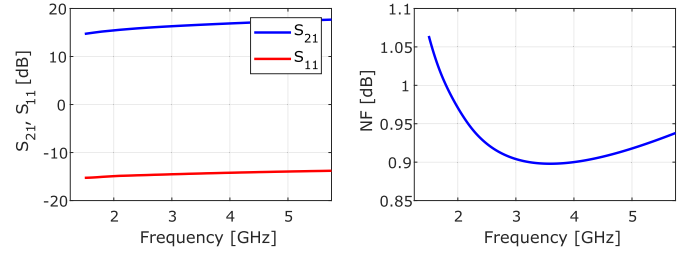


Fig. 3. Postlayout simulations of the LNA. The LNA was loaded with $60\ \Omega$ to resemble the nominal VGA input impedance and the voltage gain of the LNA is then 6.8 dB larger than S_{21} .

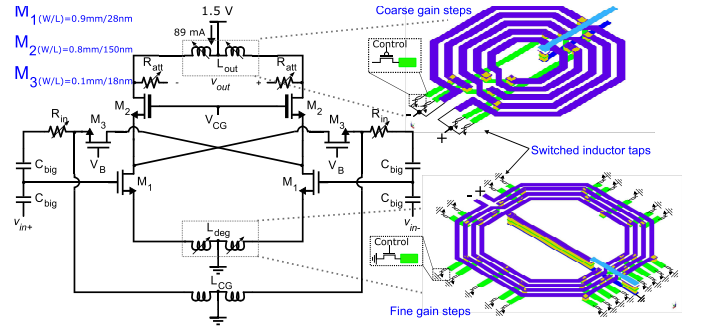


Fig. 4. Schematic of the proposed VGA.

impedance interface, the degradation in NF is then just 0.13 dB at the lowest operating frequency of 1.5 GHz. This degradation becomes negligible at higher frequencies as the parallel resistance of the inductor then increases. Simulations of the LNA gain, input matching, and NF are shown in Fig. 3. As can be seen, the LNA is well matched, the NF is below 1.05 dB, and the gain is nearly flat over the frequency range of interest.

B. VGA

The requirements on the VGA include wide BW, low NF, high linearity, and low input impedance, at different gain settings. An amplifier architecture based on a CS-CG input stage with inductive source degeneration is attractive thanks to its high linearity and low NF (see Fig. 4). An additional input stage with cross-coupled CG devices (M_3) is used to ensure wideband low input impedance over gain settings. The LNA output impedance is high at the center frequency, and when connected to the VGA, its low input impedance is therefore needed to obtain wide signal-chain BW. A programmable (switchable) input resistance (R_{in}) is also connected in series to provide more control of the VGA input impedance. A programmable inductor (L_{out}) is used as output load and for gain control, and also, the degeneration inductor (L_{deg}) is programmable. An attenuating programmable resistor (R_{att}) is used to isolate the following blocks (equalizer and output buffer) from the VGA at low gain settings.

The wide BW requirement is fulfilled by tuning L_{deg} to resonate with the parasitic capacitance at M_1 source at the same frequency as L_{out} resonates with the capacitance at the output node, including the input capacitance of the following blocks. The result is maximum source degeneration at the frequency where the output impedance is highest and reduced

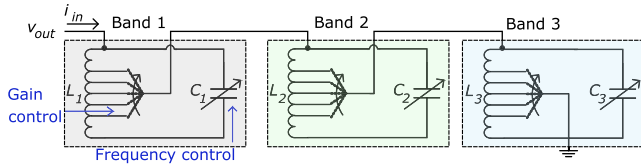


Fig. 5. Simplified schematic of the proposed three concurrent bands' equalizer (showing one side of the differential circuit).

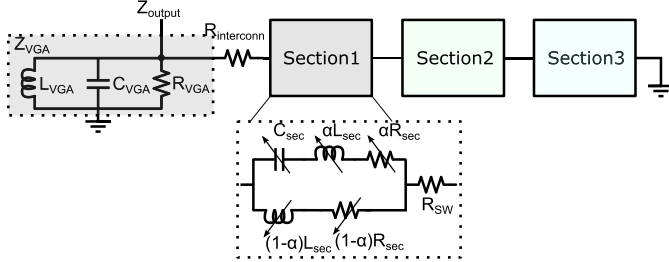


Fig. 6. Model used in the equalizer analysis.

degeneration and therefore higher effective transconductance gain at lower and higher frequencies. Given that the degeneration impedance is larger than $1/g_m$ and the two programmable inductors have similar Q value, this results in a wide BW voltage gain. The linearity is high thanks to increased supply voltage and low dc voltage drop over the inductors.

To obtain programmable gain with low noise, we use variable tuned degeneration and load impedances, where inductors L_{deg} and L_{out} are programmable using switched taps. The taps in L_{deg} are selectively connected to signal ground to program the degeneration inductance. This is limited to fine gain tuning to maintain high BW, while coarse steps are implemented in L_{out} where taps are shorted together to limit the output impedance and maintain high BW. Two differential taps are implemented in L_{out} and 11 in L_{deg} , of which two are spares.

C. Equalizer

A simplified schematic of the proposed equalizer is shown in Fig. 5. Three concurrent frequency bands are supported by connecting three LC resonators in series. A programmable switched capacitor bank is used in each resonator to tune its frequency, and an inductor with five switched taps is implemented to control the gain in each band, enabling orthogonal frequency and gain tuning with a fully passive equalizer structure. The output impedance of the VGA is narrow-band; however, thanks to large inductance, the VGA output impedance is still high at the highest frequency of interest (≈ 6 GHz). In this work, the VGA output is, therefore, connected to the equalizer. This node is also connected to the output buffer, which has high input impedance. The impedance at the VGA output is thus dominated by the equalizer. An alternative to the equalizer is to use three parallel VGAs, each tuned to one of the three wanted passband frequencies. Compared to the equalizer, this approach would have increased noise, power consumption, and power combining and splitting overhead.

To perform an analysis of the proposed equalizer, a circuit model is shown in Fig. 6. Each section of the equalizer

is modeled by an RLC circuit, where the tapping factor $\alpha \leq 1$ models the amount of tapping of the inductor L_{sec} of that section. Each section is modeled as a series resonator αL_{sec} , C_{sec} , and αR_{sec} in parallel with the remaining finite- Q inductance $(1-\alpha)L_{sec}$ and $(1-\alpha)R_{sec}$, and the loss associated with the inductor tap switch is modeled by R_{sw} . The output impedance of the VGA is modeled as a parallel RLC resonator L_{VGA} , C_{VGA} , and R_{VGA} , and the wires connecting the VGA to the equalizer are modeled as a resistor $R_{interconn}$. The impedance seen at the output of the VGA, Z_{output} , becomes

$$Z_{output} = Z_{VGA} \parallel (R_{interconn} + Z_{section1} + Z_{section2} + Z_{section3}) \quad (2)$$

where $Z_{section}$ is the impedance of the respective section. Tapping an inductor connects a portion of the inductor L_{sec} and R_{sec} in series with the capacitor C_{sec} , while the inductance and resistance placed in parallel with the capacitor is reduced by the same amount. The impedance of a section can be simplified to

$$Z_{section} = R_{sw} + (1-\alpha) \left[(1-\alpha) \underbrace{\frac{R_{sec} + sL_{sec}}{C_{sec}L_{sec}s^2 + C_{sec}R_{sec}s + 1}}_{\text{parallel resonance}} + \alpha \underbrace{(R_{sec} + sL_{sec})}_{\text{parasitic series inductor}} \right]. \quad (3)$$

As can be seen, the resonance frequency is at $\omega_o = (2\pi(L_{sec}C_{sec})^{1/2})^{-1}$ when $\alpha = 0$. Increasing α reduces the impedance of the parallel resonator by a factor of $(1-\alpha)^2$ and introduces a parasitic series inductor scaled by $(\alpha - \alpha^2)$. Since the impedance of the parallel resonator is Q times higher than the impedance of its inductor at resonance, the impedance of the parasitic series inductor will not significantly affect the frequency or impedance of the equalizer passbands for relevant values of Q and α . The resonance frequency and the impedance magnitude of the section can then be independently tuned by C_{sec} and α , respectively.

D. TX Filter

Fig. 7 shows the TX-filter path architecture. The RF input is connected to a quadrature active mixer, having a high ohmic capacitive input impedance with low impact on the TX signal path. At baseband (mixer output), low-pass filtering is performed using programmable capacitors, and the signal is scaled in amplitude and phase rotated by a programmable active vector combiner, a phase rotator similar to [17]. After that, the signal is up-converted to RF by a second active IQ-mixer, whose output current is injected at the output of PPA1. By controlling the filter path gain and phase shift to match that of PPA1, a notch in the combined PPA1 and TX-filter path transfer function is created within the passband of the frequency-translated LPF. The filter uses an external differential LO signal at twice the desired notch frequency

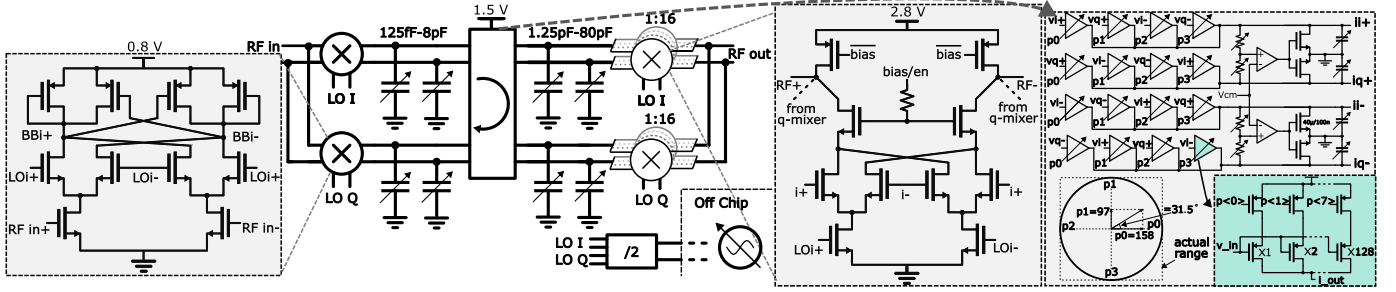


Fig. 7. TX-filter architecture, showing the mixers for frequency down-conversion and up-conversion and the IQ-phase rotator with an example phasor diagram and phasor amplitude scaling schematic.

to generate LO quadrature signals from an integrated divide-by-2 circuit. A key advantage is the flexibility to control the center frequency of the filter by changing the LO frequency. The gain of the filter path is controlled by the gain settings of the up-conversion mixer stage and the settings of the phase rotator.

1) *Quadrature LO Generation*: The quadrature LO signal for the active mixers is generated by two differential flip-flops performing a frequency divide-by-two function and a buffer stage is placed between the divider and each mixer. The differential signal amplitude at the LO input of the mixers is 800 mVpp. The current consumption of the quadrature LO generation, including buffer stages and measured at 2.8-GHz notch frequency, is 11.9 mA from the 0.8-V supply.

2) *Active Mixers*: The RF input signal is frequency down-converted to **IQ-baseband** using two active mixers clocked by LO-I and LO-Q. The mixers use an active load with self-biasing common-mode level, using diode-connected PMOS devices in parallel with cross-coupled PMOS devices (see Fig. 7). **For the frequency up-conversion from IQ-baseband back to RF, another complex mixer is used**, also clocked by LO-I and LO-Q and based on active mixers. As this mixer is connected to the output of an amplifier with an established common-mode level, PPA1, a regular PMOS current source load is used. **The up-conversion mixers are built as a set of 16 unit cells**, which can be individually enabled by turning on the bias to a cascode device (see Fig. 7). The corresponding current source loads are also then enabled/disabled to avoid injecting bias current into the PPA1 output.

3) *Phase Rotators and Low-Pass Filters*: **The IQ-baseband signal is filtered by programmable 0.125–8-pF shunt capacitors, implemented as six binary weighted cells (6 bits) connected to signal ground on each of the four IQ signals.** The filtered IQ-baseband signal is then fed to the active phase rotator (see Fig. 7). **The phase rotator is based on 16 programmable transconductance stages, organized as four quadrature stages, each controlled by a signal p_i .** The different quadrature stages provide an output signal corresponding to different baseband rotations, 0° , 90° , 180° , and 270° (see Fig. 7). **By scaling and combining the output signals in the current domain, the total output signal can be phase rotated and scaled in amplitude.** Each transconductance stage uses eight parallel binary weighted transistors (see Fig. 7), which gives a control range from 0 to 255 for the coefficients p_i ,

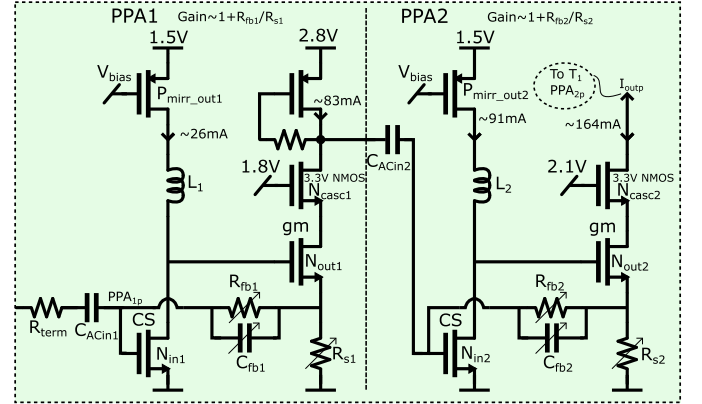


Fig. 8. Schematic of one side of differential PPA1 and PPA2.

where 0 is non-active and 255 represents the maximum amplitude. As an example, if 45° phase rotation is desired, p_0 and p_1 are activated with the same amplitude setting, and p_2 and p_3 are set to 0. **The scaled and rotated IQ-baseband output signals from the phase rotator are then each filtered by a first-order LPF realized by the output resistance of the IQ-phase rotator, which is approximately 300 Ω , and a 6-bit programmable capacitor, ranging from 1.25 to 80 pF, implemented as six binary weighted cells (6 bits) connected to signal ground on each of the four IQ signals** (see Fig. 7).

E. PPA1 and PPA2

The PPA (see Fig. 8) must provide enough output power to drive the PA input while also providing sufficient gain and a high degree of linearity over a wide BW. In addition, it should feature programmable gain functionality for TX power control. **Negative feedback topologies were favored over open-loop amplifiers.** The reason for this is that **with sufficient loop gain, both linearity and BW are improved.** The topology chosen to meet these requirements is **a cascade of two feedback current amplifiers.** The PPA has **differential input and output signals**, and **it is implemented as two independent identical single-ended amplifiers, each handling one half of the differential signal.** Being a current amplifier, the PPA has a **low input impedance**, and **to provide an impedance match with an external signal generator, the test chip uses a resistor (R_{term}) of about 50 Ω in series with each input.** Two cascaded current amplifiers, PPA1 and PPA2, were used as it **was difficult to obtain the required BW and gain in a single feedback current amplifier while also providing high linearity**

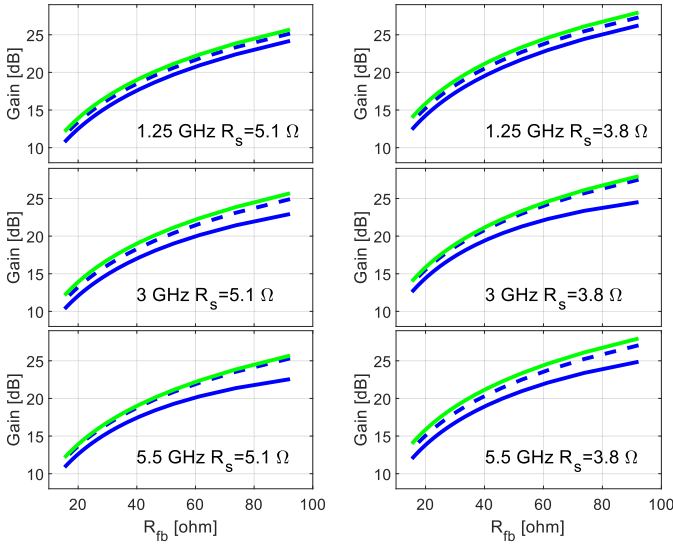


Fig. 9. Small-signal current gain at 1.25, 3, and 5.5 GHz of PPA1. Blue solid lines: Simulated. Blue dashed lines: simulated without some imperfections. Green solid lines: asymptotic gain $1 + R_{fb}/R_s$.

over the full BW. Each amplifier has $5\times$ current gain, and the cascade then provides $25\times$ (28 dB) current gain. PPA1 and PPA2 have similar designs except for the output stage load, and the component values are scaled in PPA2 to handle higher signal current levels. Each current amplifier core consists of an input CS gain stage (N_{in}) that drives a resistively degenerated (R_s) output CS stage (N_{out}), where the degeneration resistor and a feedback resistor (R_{fb}) sense the output current and feeds back a scaled copy so that the asymptotic current gain becomes $1 + R_{fb}/R_s$. To provide programmable gain with high linearity, both R_s and R_{fb} are made programmable, built from 315- Ω unit resistors (about 370 Ω , including switch R_{ON} resistance) each with a switch transistor in series. The number of resistor unit cells is different for each resistor. In PPA1, the number of unit cells for R_{fb} and R_s is 24 and 120, respectively, and for PPA2, it is instead 96 and 480. In the R_{fb} unit resistor, there is a parallel 14-fF capacitor (C_{fb}) to aid stability by creating a high-frequency zero in the feedback loop. By arranging a capacitor in parallel with the resistor in each unit cell, the frequency of the zero becomes independent of the resistor bank setting. In order to increase the high-frequency loop gain and thereby improve the linearity, an inductor (L) between the drains of N_{in} and P_{mirr_out} was added. The bias current of the PPA1 output stage is provided by a self-biasing p-channel field-effect transistor (PFET) (P_{load}) from a 2.8-V supply, and the PPA1 output signal current is ac coupled to the input of PPA2 using a capacitor (C_{ACin2}). The PPA2 output is instead connected to the primary side of the output transformer (T1) with its center tap connected to a 3-V supply, thus providing the bias current for the output stage of the second amplifier. A thick oxide (3 V) cascode device (N_{casc}) is used at the output to handle the voltage. The secondary side of the output transformer is connected to the PA input stage. The bias current of N_{in} is set using an external reference current that is mirrored to P_{mirr_out} .

Fig. 9 shows small-signal simulation results for the current gain of PPA1, compared to the asymptotic current gain

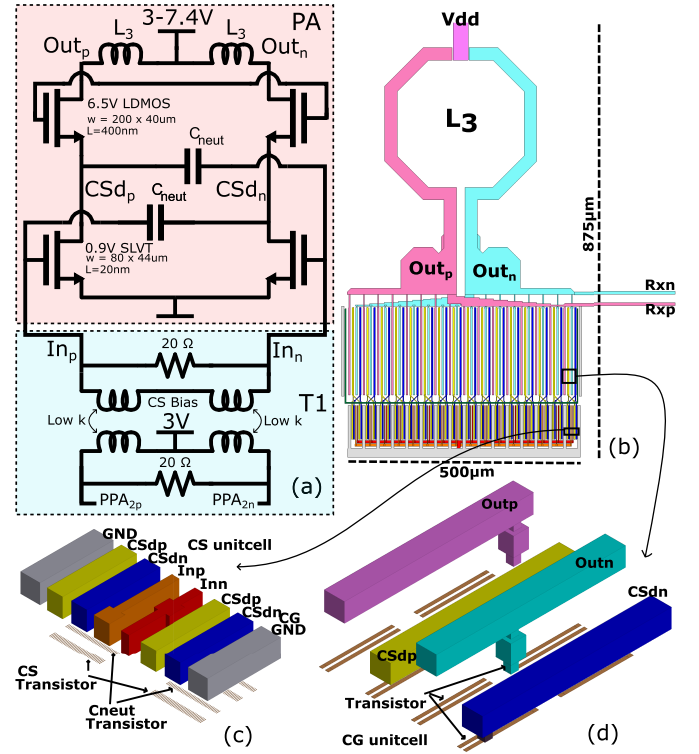


Fig. 10. (a) PA and transformer T1 schematic. (b) Overview of PA upper metal layers. (c) Upper metal layers of CS unit cell. (d) Upper metal layers of CG unit cell. The nets are color coded to ease understanding of the layout.

$1 + R_{fb}/R_s$, for two different R_s settings when sweeping the value of R_{fb} . The simulation was carried out for three different frequencies, 1.25, 3, and 5.5 GHz. A deviation between the simulated gain and the asymptotic gain is noted in all simulations. For low gain at 3 GHz, the deviation is 1.9 dB. However, the difference increases with frequency and with increased gain. To identify why the simulated gain deviates from the asymptotic one, another simulation of the current gain was performed, where only the current going into R_{fb} was regarded as input current in the current gain calculation, thus excluding the current lost into the parasitic capacitance of C_{ACin1} and C_{gs} of N_{in1} . Furthermore, the output current was taken at the N_{out1} source, and thus, current was lost in the cascade, and the N_{out1} gate was excluded. The results of these simulations are plotted as the blue dashed line in Fig. 9. This confirms that the main reasons for the difference between the simulated and asymptotic gain are parasitic capacitances in the cascode transistor of the output stage, and parasitic capacitances of C_{ACin1} together with C_{gs} of N_{in1} , in combination with limited loop gain, which deteriorates the virtual ground and exposes the input parasitic capacitance to signal voltage.

F. PA and Transformer

Fig. 10(a) shows the schematic of the PA and transformer T1, which provides a wideband low- Q low- k double resonance network between the PPA and PA. When $k \cdot Q = 1$, critical coupling occurs, yielding a flat frequency response between the two resonance frequencies. The coupling factor k is lowered by a slightly reduced primary coil radius, and Q is reduced by

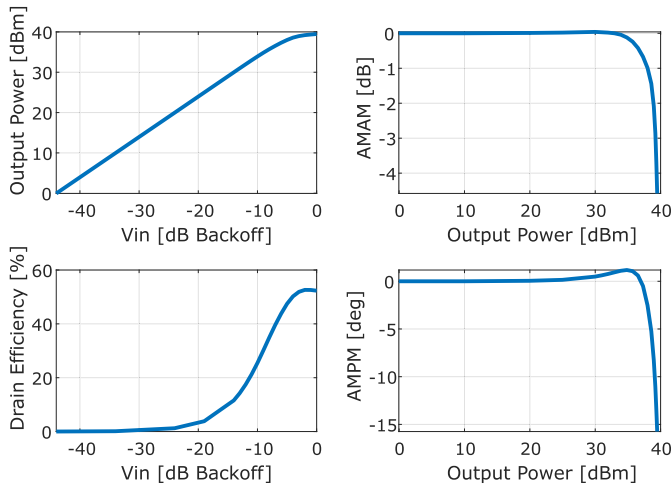


Fig. 11. PA standalone simulation results at 3.4 GHz and 7.4-V supply.

a 20- Ω parallel resistor at each side of the transformer, which also reduces the differential impedance that the PPA must drive to about 10 Ω . The downside of the proposed method is that it increases the power consumption in PPA2 as the majority of the signal power delivered by PPA2 will be dissipated in the two 20- Ω resistors that were added to achieve the required BW.

The PA uses a differential CS-CG stage biased in class AB and it has cross-coupled neutralizing capacitors for increased reverse isolation and stability. The CS part uses 0.9-V low-threshold-voltage thin-oxide transistors and the CG part features integrated 6.5-V LDMOS transistors. The LDMOS transistors are a key enabler to reach higher output power as they can handle significantly higher voltage stress than the normally used thin- or thick-oxide devices (0.9 or 1.8 V for dc). The theoretical maximum output power increase of the proposed CS-CG stage compared to a CS-CG stage with 1.8-V thick-oxide CG devices, driving the same load impedance, becomes $20\log_{10}((0.9 + 6.5)/(0.9 + 1.8)) = 8.8$ dB. The drawback of the LDMOS is primarily its lower f_t , which limits the usable frequency range, which is, however, sufficient for sub-6-GHz frequencies. To further boost output power and BW, the PA drives a differential load impedance of 12.5 Ω . The low impedance and high output power make the output current very high, and the effective g_m of the CG stages must then also be high, to reduce the load impedance of the CS stages delivering the full signal current. The CG transistor has about 20 times longer channels than the CS transistor, which, for practical implementations, makes it impossible to match the CS g_m . To avoid further increased input impedance of the CG stage, its gates are biased at differential signal ground.

Fig. 11 shows the continuous-wave (CW) simulation results for output power, drain efficiency (DE), amplitude-to-amplitude modulation (AMAM), and amplitude-to-phase modulation (AMPM), for the standalone PA at 3.4 GHz with 7.4-V supply.

To investigate the voltage stress of the PA transistors when amplifying a 5G NR orthogonal frequency-division multiplexing (OFDM) signal with an average output power of 32 dBm, a 6-dB PAR value, and, consequently, a peak signal power

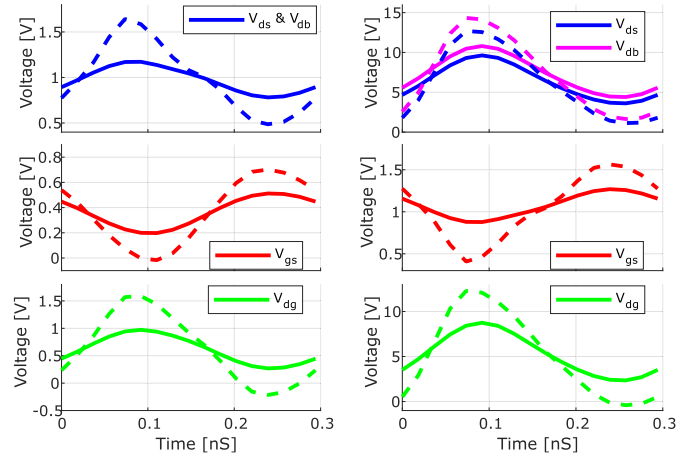


Fig. 12. Voltage levels for the PA transistors at 3.4 GHz with a supply voltage of 7.4 V and with the bulk node biased to the ground potential. Solid lines: average power level (32 dBm). Dashed lines: peak power level (38 dBm). Left: CS transistor, Right: CG transistor.

of 38 dBm, two CW-tone simulations were performed. The first simulation was configured to have an output power of 32 dBm and represents the average signal level. The second simulation was instead for an output power of 38 dBm and represents the peak signal level, which occurs about 2% of the time. To ensure that all voltages in the PA are within a reliable operating region of the devices, Fig. 12 shows the time-domain plots for V_{ds} , V_{db} , V_{gs} , and V_{dg} for the CS and CG stages of the PA. All the voltage levels are within the safe operation region according to the process design kit (PDK).

The very high current levels create major PA layout challenges, and Fig. 10(b)–(d) visualizes the measures taken to manage them. Fig. 10(b) shows an overview of the upper thick metals of the PA, where inductor L_3 is implemented using all the three thick top metal layers, two copper layers, and one aluminum, with vias between, dimensioned to reliably conduct the high dc current and provide low RF losses. The trace width is 25 μm and the diameter is 305 μm , providing a simulated inductance of 488 pH and a Q value of 20 at 2.7 GHz and 100 $^\circ\text{C}$. The effective parasitic parallel resistance of the inductor then becomes about 170 Ω , but due to the low output impedance, this only results in losses of 0.3 dB. Due to the high output power levels, however, this still corresponds to almost 0.5 W of loss at peak PA output power. The PA output resonates at 3 GHz with a Q value of 1.3 when loaded by the 12.5- Ω load. For the dc current, the center tap of L_3 is critical, and its width is increased to 30 μm . According to PDK electromigration reliability calculations, there is then a factor of 2 in the margin when operating at 120 $^\circ\text{C}$, with an ON-time duty cycle of 30% for ten years, with maximum modulated output power. Fig. 10(c) and (d) shows the CS and CG transistor unit cells, respectively. The differential CG stage uses 96 unit cells and is laid out slightly less dense than possible from design and electromigration rules, to spread its heat over a larger area. To handle the dc current, the CG stage has 32 parallel 3- μm traces, i.e., 16 for each side of the amplifier. The differential CS stage has 384 unit cells and 64 parallel 2.4- μm traces. The CS unit cells also contain

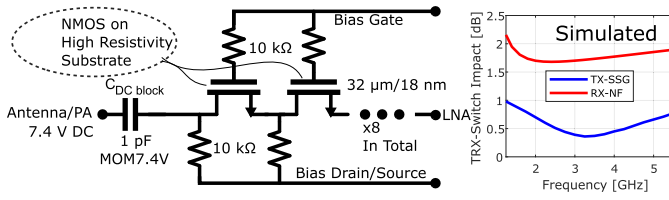


Fig. 13. Left: simplified schematic of the TRX-Switch. Right: simulated impact of TRX-Switch, loss of TX small-signal gain, and RX NF increase without the PA connected.

the neutralizing capacitors, implemented as OFF-state biased transistors with reduced width. The 18 ground and 12 supply pads are used to supply the PA with dc power. The complete PA, including all its pads, occupies 1.3-mm² die area.

The PA supply is turned on at all times to avoid settling issues related to quickly ramping up/down the supply voltage, but to save power, the PA must be turned off when operating in the RX mode. This is accomplished by pulling the PA CS gate bias to the ground and simultaneously reducing the CG gate bias voltage enough to avoid a too high dc voltage across the thin-oxide CS device. All PA bias voltages are derived internally on the chip and can be digitally controlled.

G. TRX-Switch

Fig. 13 (left) shows a schematic of one side of the differential TRX-Switch, implemented as a single-pole–single-throw (SPST) switch. It consists of a dc block capacitor followed by eight stacked thin-oxide NMOS transistors placed on a high resistivity doped region of the substrate. In the TX mode, all switches are open, i.e., non-conducting, and since the PA and antenna are connected to the same side of the TRX-Switch, it should then ideally not affect the signal. However, parasitic capacitances of the switch will still load the PA, and their finite Q will cause RF signal losses. Furthermore, the increased capacitance will degrade the PA BW, even if the PA output is re-tuned with smaller inductance to maintain the center frequency. Fig. 13 (right) shows the simulated impact on the TX small-signal gain. The TRX-Switch must handle the large output signal from the PA, which despite the low impedance still becomes up to $V_{\text{peak}} \approx 6.4$ V at each side of the differential signal, which, for reliability, is distributed across eight transistors. Fig. 13 (right) shows the simulated insertion loss in the TX mode and NF increase in the RX mode from 1.25 to 5.5 GHz. The TX-to-RX isolation of the switch is simulated to more than 37 dB for the whole frequency range.

V. PACKAGE AND PCB STACK UP

The chip was designed for a flip-chip chip-size package. Fig. 14 (left) shows a cross section of the die and the ball grid array (BGA) package. The backside of the chip is exposed to allow for a direct heat sink contact. The package has two build-up layers around a thicker core, and strip lines can be implemented on layer 2 (L2) or layer 5 (L5), as indicated in Fig. 14. The 12.5-Ω balanced signal is routed through the package, and the low impedance makes inductances in the via and bump transitions problematic. This is different from using more conventional impedance levels, as the capacitive loading of the bump and via pads then instead have more

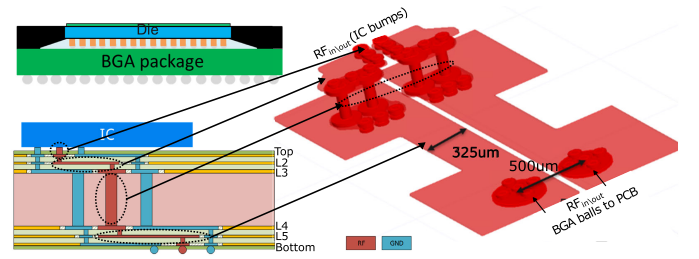


Fig. 14. Top left: BGA build up. Bottom left: package stack up. Right: TRX signal routing from the IC to the package.

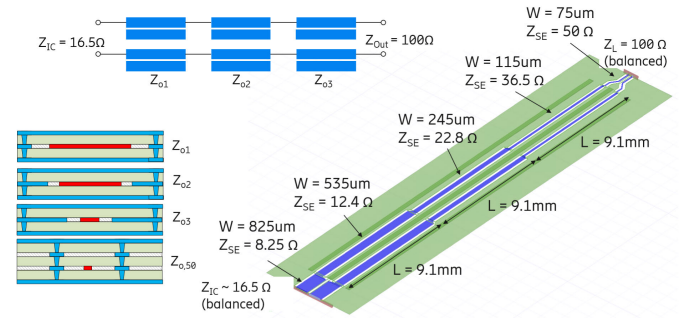


Fig. 15. Overview of PCB stack up and the implemented three-stage TLT.

impact on impedance matching quality. In the package, a low characteristic impedance is achieved using a wide transmission line structure in layer 5, as shown in Fig. 14 (right). Copper shapes were also added to provide capacitances resonating with the inductances in the package vertical transitions.

A five-layer PCB was used, with any layer connectivity, which allowed for micro-vias to fan-out signals and supplies from the BGA. The BGA converts the chip 12.5-Ω differential interface to 16.5 Ω on the PCB. This is converted to a 100-Ω differential impedance by the TLT, which is implemented on the PCB using three differential transmission line sections [13]. Each section is 9.1 mm and the characteristic impedance is progressively increased, the first section having 12.4-Ω single-ended impedance, the next having 22.8 Ω, and the last having 36.5 Ω (see Fig. 15). The signal path from the die pads all the way to the 100-Ω interface on the PCB was simulated and optimized in Ansys HFSS EM simulation software to obtain good matching and low insertion loss over the frequency range. The RX test input features a very similar signal path and the same PCB TLT is reused. The overall simulated loss from die bumps to the subminiature version A (SMA) connector is below 0.8 dB at 3 GHz (<0.5 dB for the TLT alone), which is also in close agreement with measurements of a PCB test structure that, however, does not include effects of the package and on-chip bumps.

A. Thermal Considerations

To improve the cooling of the die, its backside was exposed at the upper side of the flip-chip package. A thin layer of liquid metal, which acts as a thermal interface material (TIM), was used between the exposed silicon die and a copper heat sink. Heat from the PA can then effectively be conducted through

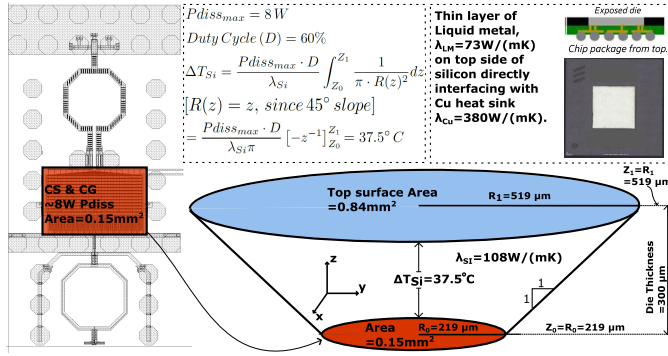


Fig. 16. Left: PA layout indicating the major heat dissipating area. Top middle: analytical expression for thermal resistance based on model assumptions. Top right: exposed die package illustrated from the side and photographed from the top. Bottom right: heat transfer model with basic assumptions indicated.

the silicon chip to the back side and then via the TIM directly to the heat sink. To reduce the thermal resistance of the die, its thickness was reduced to 300 μm . The used liquid metal TIM has a very high thermal conductivity of $\lambda_{\text{TIM}} = 73 \text{ W/mK}$, which can be compared to a grease-based TIM from the same manufacturer having 12 W/mK [18], [19]. The liquid metal is also expected to provide a much thinner TIM layer than grease. Having reached the heat sink, the heat will effectively spread evenly inside it due to the very high thermal conductivity of copper ($\lambda_{\text{Cu}} = 380 \text{ W/mK}$). A fan was mounted on the heat sink to improve the heat flow to the surrounding air and reduce its temperature.

The PA bias is digitally controlled by DACs that set the gate voltage of both the CS and the CG stages. To be able to operate the PA in a pulsed mode, mimicking a 5G time-division duplex (TDD) pattern, an external pulse control signal was added to the chip. This provides a mean for fast ON/OFF switching of the PA, pulling the CS gate bias to ground to turn the PA OFF while reducing the CG gate bias voltage enough not to damage the CS transistor, while the PA supply voltage remains high. This enables the PA to operate with high instantaneous but reduced average power dissipation. At maximum output power, for modulated signals, the PA delivers 32.1 dBm with a DE of 16.4%, which results in an on-chip power dissipation of more than 8 W heating the chip. The main part of this power dissipation occurs in the rather limited layout areas of the CS and CG stages.

Fig. 16 shows a model of the PA heat transport with the measures taken to improve the cooling. Fig. 16 (left) shows a layout of the PA indicating where the majority of the heat is dissipated. Here, we assume that all heat is conducted through the silicon die to the heat sink, ignoring other heat paths, for instance through the BGA and PCB. The model further assumes a heat circle on the chip surface with the same area as the CS-CG rectangle in the layout, and as the heat is conducted through the silicon, it linearly and evenly spreads in the horizontal directions until it reaches the liquid metal-covered surface, which is in direct contact with the cold heat sink. With those assumptions, the thermal resistance of the path through the chip can be found by integrating the thermal resistance of thin cylinders with gradually increasing radius (from R_0 to R_1), along the z -axis, and the delta temperature

(ΔT_{Si}) through the chip can then be estimated by the average dissipated power multiplied by the thermal resistance.

VI. EXPERIMENTAL RESULTS

Fig. 17 shows a die micrograph, the PCB, and the test setup for TX and RX measurements. Including bumps, bias circuits, digital control interface, and auxiliary blocks, the TRX part of the chip occupies 6.6 mm^2 in a 22-nm fully depleted silicon on insulator (FD-SOI) process. External hybrid couplers (HCs) were used as baluns for input and output signals in both TX and RX measurements. RX bias voltages were generated by resistor-2resistor (R-2R) DACs, with inputs from a serial to parallel interface. The DACs were supplied by the local 1.5-V supplies of the LNA, VGA, and output amplifier. A differential cascode amplifier loaded by an inductor in parallel with an on-chip 100- Ω resistor was used as an RX output buffer to drive the 50- Ω instruments. The test input was used in the RX measurements eliminating the effect of switch losses. Both TX and RX losses external to the packaged chip, due to power splitters, attenuators, HCs, cables, and the TLT, are measured and de-embedded. In TX, the losses in the package, however, which from simulations are estimated to be 0.2–0.3 dB between 1.25 and 5.5 GHz, are not de-embedded, which at peak TX output power corresponds to approximately 300 mW of signal power loss.

Both RX and TX small-signal measurements, including NF, S-parameters, and gain, as well as RX large-signal measurements, including intermodulation distortion and compression, and TX AMPM, were performed using the performance network analyzer-series (PNA-X) network analyzer from Keysight (see Fig. 17). The TX P_{out} versus input power (P_{in}) characteristic and AMAM modulation were measured with an R&S SMW200A vector signal generator together with an Agilent Technologies MXA Signal Analyzer. An R&S SMW200A vector signal generator and a Keysight M9384B vector signal generator were used to generate modulated input signals, either using predefined instrument available 3rd generation partnership project (3GPP) test signals or by uploading IQ signal files to the instruments. A Keysight UXR0502A Real Time Oscilloscope with Pathwave Vector Signal Analysis (89600 VSA) software was used to demodulate and analyze the modulated output signals.

A. RX Results

All RX measurements were performed by feeding the signal to the RX input port (RF test_{in}) (see Fig. 1). Small-signal measurements, including gain, input match (S_{11}), and NF at different gain settings, are shown in Fig. 18(a). The equalizer was disabled by shorting all inductor taps together, and its impedance is then dominated by the parasitic capacitances of the switches and capacitor banks, slightly reducing the measured RX BW. The maximum gain exceeds 25 dB across the 3-dB BW (1.55–4.8 GHz) and S_{11} is better than -12 dB across the range. At maximum gain setting, an excellent 1.3-dB NF, despite interfacing with 6.25 Ω , is achieved. Backing off the RX gain by 18 dB using the VGA gain programming increases NF by only 6 dB. These results are achieved thanks to the LNA and VGA architecture.

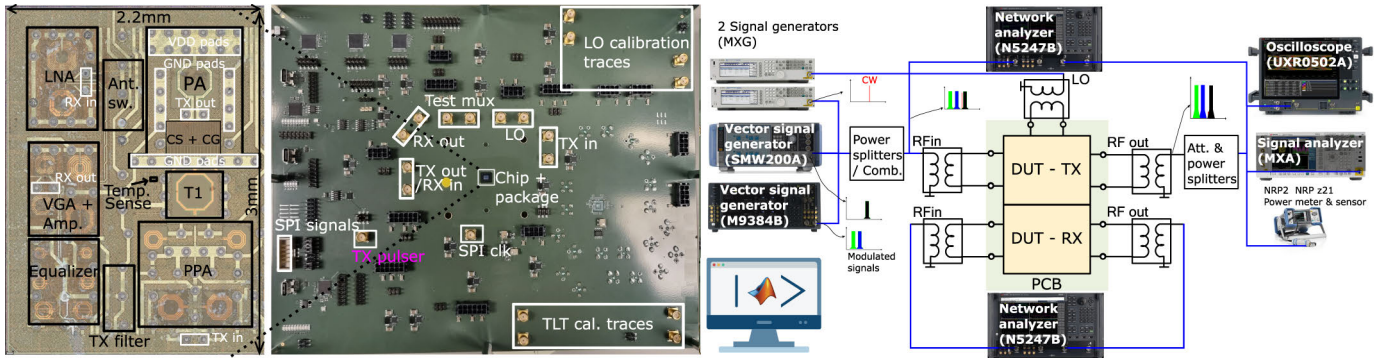


Fig. 17. Left: chip photograph of the complete TRX front end. Middle: PCB with the mounted chip in package near the middle. Right top: simplified TX measurement setup. Right bottom: simplified RX measurement setup.

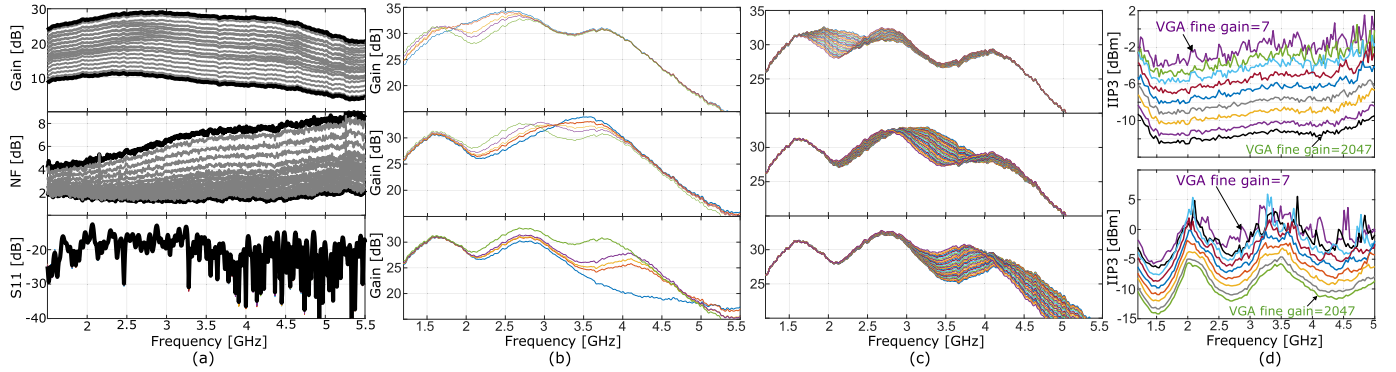


Fig. 18. (a) Front-end gain S_{21} (top), NF (middle), and S_{11} (bottom) at different VGA coarse and fine gain settings. (b) Front-end gain when different inductive taps in band 1 (top), band 2 (middle), and band 3 (bottom) are selected, while taps in other bands are set at maximum gain. (c) Front-end gain when capacitor banks in-band 1 (top), band 2 (middle), and band 3 (bottom) are swept, while capacitor banks in other bands are fixed. (d) IIP3 of the proposed front end with maximum coarse gain and different fine gain settings when the equalizer is disabled (top) and enabled (bottom).

To test the equalizer functionality, the inductor tap of each resonator is varied, while other resonators are fixed [see Fig. 18(b)], applying a gain variation to each frequency band separately. The first two band frequencies are close to each other limiting the individual gain tuning to around 5 dB in the second band and even less in the first band. The simulated value is about 10 dB, the difference being attributed to long traces in the layout reducing the quality factor (see Fig. 6), but the achieved filtering is still useful considering that it relaxes the dynamic range by 6 dB corresponding to a reduction of 1 bit in the ADC dynamic range.

Sweeping the capacitor banks with the taps fixed, the frequency tuning characteristic of the equalizer for each band is shown in Fig. 18(c). As can be seen, the frequency of each band can be separately tuned with minor effects on the neighboring frequency bands. In between bands, a potential blocker will be shunted to the ground by a low equalizer impedance, improving the overall linearity of the RX as will soon be seen.

The linearity of the front end was evaluated in two-tone tests, where the third-order intercept point (IIP3) for maximum gain coarse setting and swept fine gain setting was investigated. The tone spacing was 100 MHz, and measurements were performed both with the equalizer enabled and disabled [see Fig. 18(d)]. The input compression point was measured to ~ 10 dB below the IIP3. As can be seen in Fig. 18(d), at different fine gain settings, the IIP3 varies from ~ -12 to ~ -3 dBm

with the equalizer disabled. With the equalizer enabled, 21- Ω R_{att} was chosen in the VGA, and between the bands, up to 12-dB improvement in IIP3 was then observed compared to in-band IIP3. When compared to the wideband case with the equalizer disabled, up to 8-dB IIP3 improvement is achieved. The measurements clearly indicate that considerable linearity benefits can be achieved with the proposed equalizer.

B. TX Results

The TX measurements were performed with the PA in a pulsed mode, resembling operation in a TDD system, but with prolonged ON/OFF-times of $T_{ON} = 50$ ms and $T_{OFF} = 450$ ms, i.e., with a TX duty cycle of 10%. The PPA, however, was always ON during measurements. The measured power levels, for all results, were calibrated using an R&S NRP-Z21 Power Sensor. PA supply current was measured using an Agilent 1147A 50-MHz current probe, for efficiency calculations. All measurements, both CW and modulated, were carried out with identical TX settings. PPA1 then consumes 51/274 mW from 1.5-/2.8-V supplies, and PPA2 uses 256/1.6 W from 1.5-/3-V supplies, i.e., the total PPA power consumption is 2.2 W.

1) *CW Measurements*: Fig. 19 shows the measured CW performance at 3.4 GHz, with record high output power for both 6.5- and 7.4-V supply (37.7 and 38.1 dBm), high AMAM and AMPM linearity, and a DE as expected for a class AB PA. Fig. 20 shows P_{sat} over more than two octaves of BW (1.25–5.5 GHz), measured at -30 °C, 25 °C, and 65 °C ambient

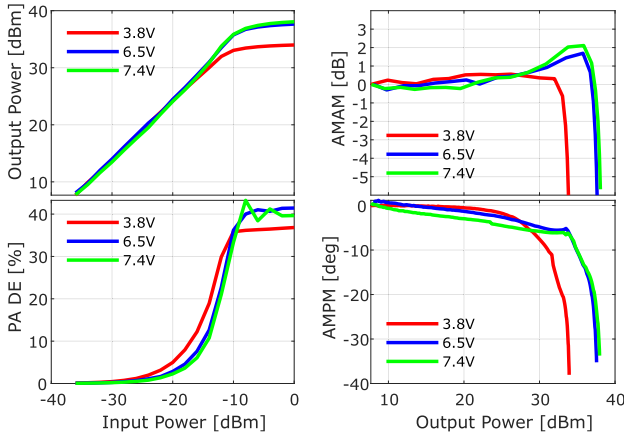


Fig. 19. Measured output power and DE versus input power and AMAM and AMPM versus output power at 3.4 GHz.

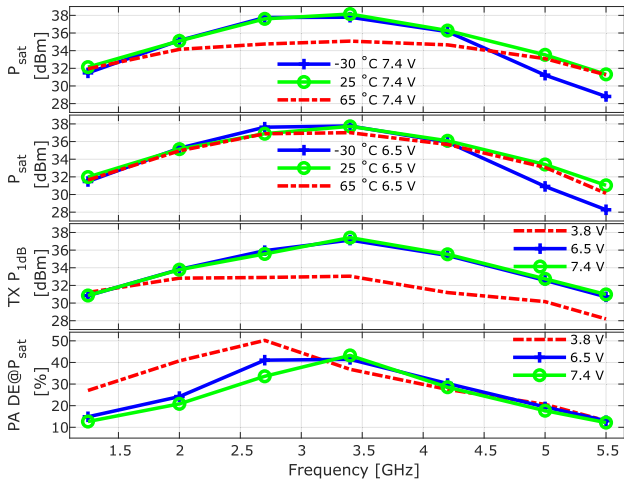


Fig. 20. P_{sat} versus frequency for 7.4- and 6.5-V PA supply, for -30°C , 25°C , and 65°C ambient temperature, and TX $P_{1\text{dB}}$ versus frequency at 25°C , and PA DE at P_{sat} versus frequency at 25°C .

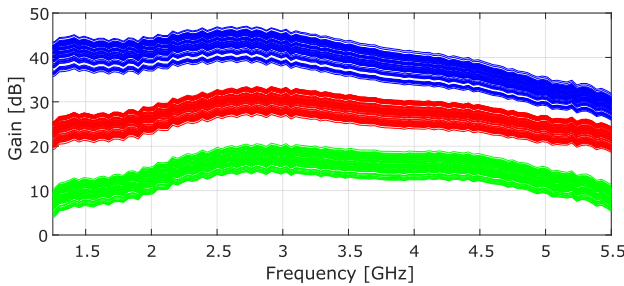


Fig. 21. Measured TX gain for 64×3 selected PPA gain settings.

temperature for two supply voltages (7.4 and 6.5 V). At 25°C , the DE at P_{sat} and the 1-dB compression point ($P_{1\text{dB}}$) are also shown for three PA supply voltages (3.8, 6.5, and 7.4 V). Fig. 21 shows the TX gain frequency characteristics for 64 fine gain settings around three coarse gain settings. This represents a small selection of all possible gain settings of the PPA, which offers multiple possibilities to cover the full gain range with a fine resolution.

2) *Modulated Measurements*: In the modulated measurements, the values stated for in-band output power, DE,

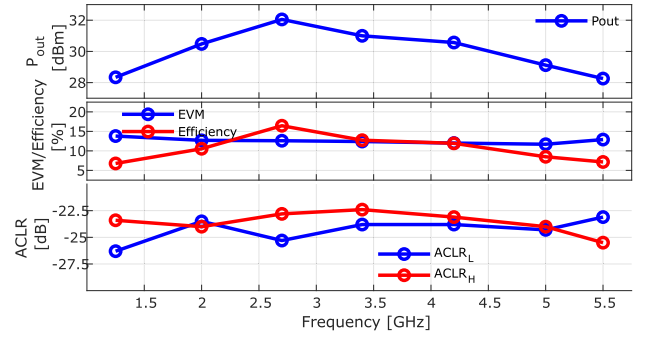


Fig. 22. 6.5-V PA supply and 100-MHz OFDM 16-QAM.

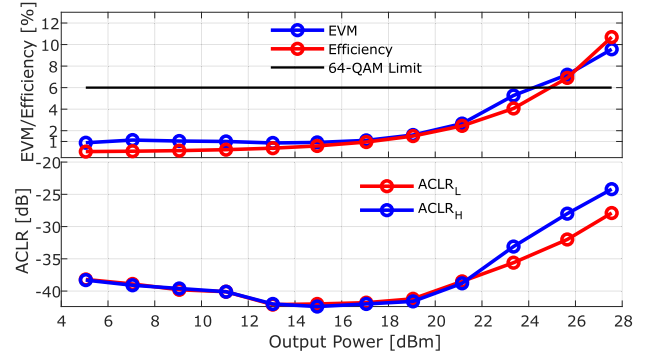


Fig. 23. Measured performance using 3.8-V PA supply, modulated 100-MHz BW 5G NR 64-QAM-OFDM signal at 2.7 GHz with PAR of 12.7 dB, sweeping input signal power with fixed gain settings.

and adjacent channel leakage ratio (ACLR) are average values for the modulated signal. For error vector magnitude (EVM), the rms value for all subcarriers in all modulated carriers is shown. No de-embedding of EVM contributions from the test setup was done. Fig. 22 shows the measured output power, EVM, DE, and ACLR for a 100-MHz 16-QAM 5G NR OFDM signal at different carrier frequencies, and as can be seen, more than 32-dBm average power at 2.7 GHz can be achieved with an EVM of 12.6% and a DE of 16.4%.

Fig. 23 shows TX EVM and ACLR, and PA DE, versus output power level for a 100-MHz 5G 64-QAM NR OFDM signal for a reduced PA supply voltage of 3.8 V, and as can be seen at 15-dBm output power, the EVM is below 1%.

3) *Pre-Distorted Measurement Results*: An 8×100 MHz 256-QAM OFDM signal with a 9-dB PAR limitation and a sample rate of 1.96 Gs/s was used to measure the raw (non-linearized) wideband TX performance. A rather low-complexity digital pre-distorter (DPD) with a generalized memory polynomial (GMP) actuator with five memory taps, each with a fifth-order polynomial, with coefficients calculated from the raw performance, was used to pre-distort the input signal. The pre-distorted signal was sampled at 2.56 Gs/s to generate the input signal file for the signal generator, but due to instrument limitations, the effective signal BW was about 1200 MHz, which unfortunately limited the out-of-band linearization capability of the DPD. An iterative learning control (ILC) actuator DPD was also constructed to pre-distort the 800-MHz signal. A lower limit on achievable EVM of about 2% is set by the 9-dB PAR limitation of the input signal. Fig. 24 shows the measured TX output constellation diagram

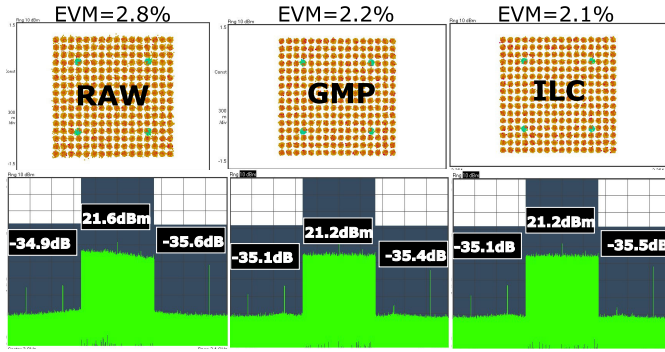


Fig. 24. Comparison of TX raw performance to GMP and ILC digitally pre-distorted input signals for a modulated 8×100 MHz BW 5G NR 256-QAM-OFDM signal at 3 GHz with PAR limited to 9 dB and 3.8-V PA supply.

TABLE II

COMPARISON OF RAW AND PRE-DISTORTED PERFORMANCE

PA Vdd V	Freq. GHz	5G-NR QAM	Mod BW MHz	Lin. Meth. dBm	Pout dBm	EVM %	ACLR Low dB	ACLR High dB
6.5	2.7	64	100	Raw	30.5	8.1	26.2	24.5
6.5	2.7	64	100	ILC	30.1	5.5	28.7	28.0
3.8	3.0	256	8x100	Raw	21.6	2.8	34.9	35.6
3.8	3.0	256	8x100	GMP	21.2	2.2	35.1	35.4
3.8	3.0	256	8x100	ILC	21.2	2.1	35.1	35.5
3.8	3.0	256	4x400	Raw	17.3	2.4	30.9	32.5

and spectrum for the raw, GMP, and ILC pre-distorted input signals. Table II summarizes and compares the performance of the raw TX with different pre-distorted measurements. In addition, to further demonstrate the wideband capability, the raw performance of a 4×400 MHz highly linear transmission of a 256-QAM signal is also included in Table II.

4) *Thermal Measurement Results*: An integrated temperature sensor was implemented on the chip rather close to the PA transistors (CS + CG) (see Fig. 17). The temperature sensor outputs a voltage through a high-impedance auxiliary test signal multiplexer. The temperature measurements started with a calibration procedure, during which all major on-chip power consumers were turned off, the ambient temperature was swept from 25 °C to 135 °C in steps of 10 °C, and the output voltage from the temperature sensor was recorded. Using the calibration results to convert measured voltage to temperature, Fig. 25(a) shows the die temperature transient for a 50-ms PA-ON pulse for different ambient temperatures. During the measurement, the PPA is always ON and its power dissipation elevates the die temperature by 22 °C from the ambient temperature. Fig. 25(b) shows the measured die temperature for different PA duty cycles. The green dashed line represents the average temperature for 60% duty cycle and is identical to the scenario investigated in Section V-A, where it was estimated that the PA temperature should be elevated by 37.5 °C. The measured average temperature is 82.5 °C, which is 35.5 °C above the ambient temperature plus the 22 °C increase due to the PPA, which is rather close given the simplifying assumptions of the model.

5) *TX Filter CW Measurements*: Fig. 26(a) shows how the TX filter notch frequency can be controlled by the LO frequency, from 1 to 6 GHz in steps of 0.5 GHz. For each

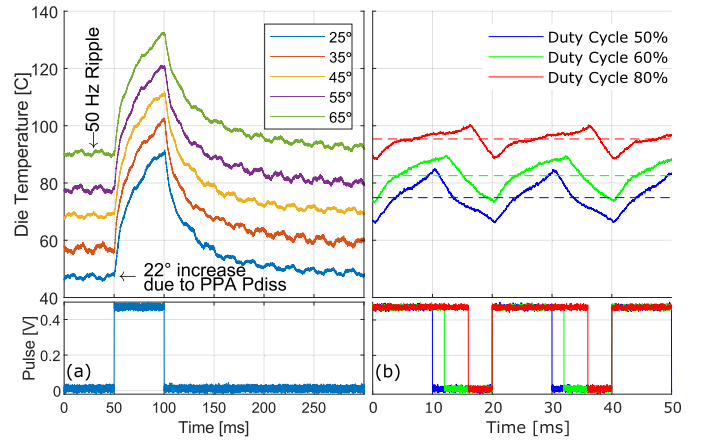


Fig. 25. Transmission of a 32.1-dBm 100-MHz BW 5G NR 16-QAM-OFDM signal at 2.7 GHz with 6.5-V PA supply. The PPA is constantly turned on and elevates the temperature by 22 °C. (a) Measured die temperature transient for a 50-ms PA-ON pulse. (b) Measurements with 25 °C ambient temperature. Solid lines: measured die temperature versus time for a 20-ms period with 50%, 60%, and 80% duty cycle. Dashed lines: average measured die temperature.

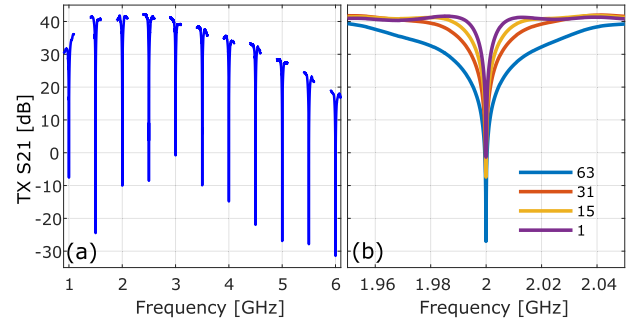


Fig. 26. (a) Measured TX notch for different LO frequencies. (b) Measured TX-notch characteristics for 4 out of 63 different BW settings.

TABLE III
RX PERFORMANCE COMPARISON

Parameter	This work		TCAS-I '17 [15]	TCAS-I '22 [21]	TCAS-II '23 [22]	TCAS-I '23 [20]
Structure	LNA	LNA+VGA	LNA	LNA	LNA	LNA
Tech. node [nm]	22	22	130	28	22	130
Configuration	Diff.	Diff.	Diff.	Single	Single	Single
NF min. [dB]	1.3 ⁽¹⁾	1.3 ⁽¹⁾	1.8	2.5	1.46	0.8
NF [dB]	1.3-2 ⁽¹⁾	1.3-2 ⁽¹⁾	1.8-3.4	2.5-3.5	1.46-1.96	0.8-1.5
Freq. [GHz]	1.5-4.8	1.5-4.8	1.2-6.6	0.02-2	2.1-5.2	6-12
IIP3 [dBm]	5 ⁽²⁾	-13- -1 ⁽⁴⁾	1.8-3.8	1.3-4.3	N.R.	N.R.
Gain [dB]	18 ⁽²⁾	29-10	14	18.5	15.7	23.3
Matched to [Ω]	6.25	6.25	50	50	50	50
S ₁₁ [dB]	<-12	<-12	<-10	<-15	<-10	<-14
I _{DC} , V _{DD} [mA, V]	83, 1.5 ⁽³⁾	172, 1.5	11, 1.2	4.1, 1	16, 1	20, 3.3

N.R. Not reported, ⁽¹⁾At RX input, TLT dembedded, ⁽²⁾Simulated, ⁽³⁾Excluding VGA, ⁽⁴⁾At max. & min. gain.

notch frequency, the TX-filter gain and phase settings are reconfigured to achieve a high level of rejection.

Fig. 26(b) shows how the notch BW can be controlled by adjusting the LPF capacitors before and after the phase rotator. As can be seen, the notch center frequency is nearly independent of the BW setting.

The linearity of the TX filter is investigated by a power sweep (see Fig. 27). The TX filter is then configured to suppress frequencies of around 2.9 GHz using a mid-range

TABLE IV
TX PERFORMANCE SUMMARY AND COMPARISON

Parameter	This Work		RFIC'20 [23]	RFIC'21 [24]	JSSC'23 [25]	JSSC'20 [26]	TMTT'21 [27]	JSSC'22 [28]	TMTT'22 [29]
Technology [nm]	22FDSOI CMOS		130 SOI CMOS	130 SOI CMOS	28 Bulk CMOS	SiGe-BiCMOS	250 GaN	250 GaN	CGHF40010F CGHV27030S
TRX-Switch	yes		No	No	No	No	No	No	No
Pwr.Combining	1		1	2	8	2	DEPA	Doherty	LMBA
Gain Control [dB]	> 35		No	No	No	No	No	No	No
No. of Stages	3		2	2	1	3	1	1	1
Area [mm ²]	3		2.21	1.92	1.4	2	10.2	7.8	7200 ⁽⁵⁾
Vdd [V]	7.4	6.5	5	5	1,1 / 2,2	5	28	28	50, 28, 2.7
Frequency [GHz]	3.4⁽³⁾	2.7	2.3	2.44	2.9	5	3.5	4.4	3.5
P_{sat} [dBm]	38.1	36.9	35.8	35.1	36.1	33	41.7	39.5 ⁽¹⁾	48
P_{1dB} [dBm]	37.4	35.6	35 ⁽¹⁾	34	n/a	32	36 ⁽¹⁾	32 ⁽¹⁾	36 ⁽¹⁾
DE_{max} [%]	39.7	41.4	N.R.	N.R.	N.R.	N.R.	56	55 ⁽¹⁾	55 ⁽¹⁾
PAE_{max} [%]	34.9⁽²⁾	35.4⁽²⁾	57	53	33.6	35.5	N.R.	41.2 ⁽¹⁾	N.R.
Power Gain [dB]	44.2	45.5	30	29.5	n/a	31.8	10 ⁽¹⁾	9.5 ⁽¹⁾	8 ⁽¹⁾
Modulated meas									
Signal type	OFDM	OFDM	LTE QPSK	WiFi	QAM LTE	802.11n WLAN	OFDM ⁽⁴⁾	64-QAM ⁽⁴⁾	OFDM ⁽⁴⁾
BW [MHz]	100	100	20	40	20	40	100	100	900
Constellation	16	16	QPSK	1024	64	16	N.R.	64	N.R.
Pout [dBm]	32.6	32	31	22.5	29.4	25.5	33.8	32.5	37.8
EVM [dB]	-17.6⁽²⁾	-18⁽²⁾	N.R.	-43	-25	-30	N.R.	-23.7	N.R.
PAE [%]	13.5⁽²⁾	13.6⁽²⁾	42.6	16	18.5	16.3	45.5	42.3	44 ⁽¹⁾

N.R. Not reported, n/a Not applicable, PAE Power added efficiency, ⁽¹⁾Graphically estimated, ⁽²⁾Complete TX, ⁽³⁾2.7 GHz for mod. meas, ⁽⁴⁾Including DPD, ⁽⁵⁾Board area.

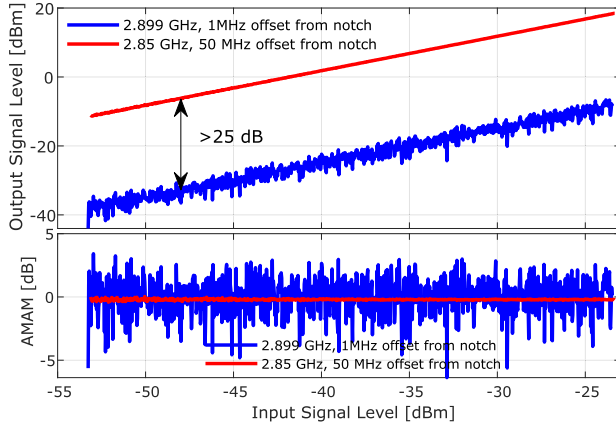


Fig. 27. Top: measured TX output power versus input power for tone at 1- and 50-MHz offset from the notch frequency. Bottom: AMAM for the output signal of the top plot.

setting for the filter capacitors [corresponding to “31” in Fig. 26(b)], and the output power versus input power is measured for 1- and 50-MHz offset from 2.9 GHz. As can be seen, the TX filter (and the TX) is very linear for input signal levels ranging from -58 to -28 dBm both for 50-MHz offset and 1-MHz offset from the notch.

6) *TX Filter Modulated Signal Measurements*: The LO signal leakage at the TX output is measured in Fig. 28(a), and due to the high TX gain, the level is -9 dBm. It has been concluded from measurements that the dominant cause for LO leakage is a mismatch in the down-conversion mixer. A suitable technique to reduce it would then be to introduce DACs injecting small dc currents to cancel the error at the I and Q output signals of the mixer, similar to [1], where LO leakage was calibrated in an I and Q TX.

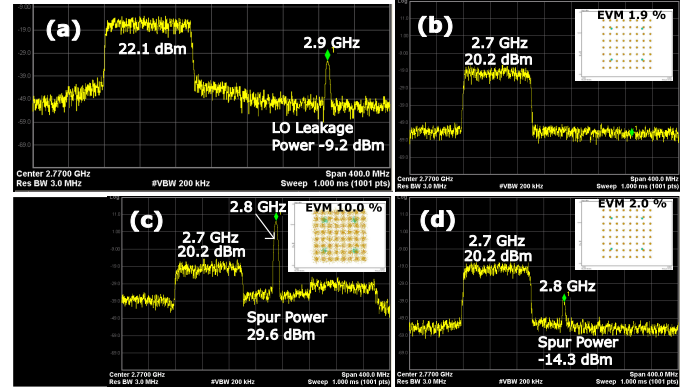


Fig. 28. Transmission of a 20.2-dBm 100-MHz 64-QAM OFDM 5G NR signal at 2.7 GHz with constellation diagram as an inset. (a) Measured TX-filter LO leakage. (b) TX filter deactivated, no CW tone. (c) TX filter deactivated, while simultaneously transmitting a 29.6-dBm CW tone at 2.8 GHz. (d) TX filter activated, and same input CW tone as for (c), but now suppressed by 43.9 dB.

In Fig. 28(b), a reference measurement was performed for a 100-MHz 64-QAM OFDM 5G NR signal transmission at 2.7 GHz with an average channel power of 22.2 dBm and an EVM of 1.9%. In Fig. 28(c), a large CW-tone signal resulting in an output power of 29.6 dBm at 2.8 GHz is also input to the TX. The demodulated OFDM signal now shows a significantly degraded EVM of 10.0%. To demonstrate the TX-filter capability of handling very large signals, without negatively affecting the desired signal, the TX filter was activated and configured to suppress the large CW tone in Fig. 28(c) scenario, which corresponds to a large spurious signal entering the TX. The TX filter suppresses the CW tone at 2.8 GHz by >43 dB [see Fig. 28(d)], and as the PA does not have to handle the large CW tone, the EVM of the OFDM

TABLE V
TX FILTER PERFORMANCE SUMMARY AND COMPARISON

Parameter	This Work	JSSC'13 [30]	JSSC'23 [31]	MWTL'24 [32]
Architecture	Freq. Trans. LP.	Active N-Path	Quasi-Elliptical N-Path	N-Path
Configuration	Notch	BP	BP	BP + Balun
CMOS node [nm]	22FDSOI	65	65	65
Freq. [GHz]	1-6	0.1-1.2	1-5	0.1-1
BW [MHz]	1-10⁽¹⁾, 0.4-4⁽²⁾	8	5-80	30
Filter Order	1	6	3	2
Rejection [dB]	> 43	59	24	16
Pow. Cons. [mW]	206	18-57.4	40-167	12.2
Area [mm ²]	0.14	0.27	4.6	0.32

BP Band-pass. ⁽¹⁾For 20 dB rejection. ⁽²⁾For 30 dB rejection.

signal transmission is reduced down to 2.0%, close to the reference measurement of 1.9%.

C. Comparison Tables

The RX performance is summarized and compared to state of the art in Table III. As can be seen, the IIP3 is excellent, and the proposed work even shows excellent NF compared to works that have an input matched to 50 Ω . The LNA presented in [20], however, uses a technology with high inductor Q and is matched to 50 Ω , enabling a sub-1-dB NF.

Table IV compares the TX to published high-power state-of-the-art PAs. The comparison includes three integrated CMOS PAs, one integrated SiGe-BiCMOS PA, and two integrated and one discrete GaN PA. When compared to the CMOS PAs and the SiGe-BiCMOS PA, the proposed TX reaches the highest reported output power in the frequency range, even though fabricated in a smaller feature size technology than the other circuits. All the GaN PAs deliver high output power with excellent efficiency. However, they all suffer from low gain, low covered BW, large chip area, and poor linearity. All the modulated measurements for the GaN PAs are therefore dependent on having a highly capable DPD linearizing the output signal. Furthermore, since they are not integrated in the TRX circuit, they will all have to be added as an external component.

Table V compares the TX filter to published frequency-translating RF filters in CMOS technology. The filters in the comparison are N -path bandpass filters for RX front ends, so they have different use cases than our notch filter for a TX. A direct comparison therefore becomes difficult, but the frequency of operation combined with rejection and chip area is strong points for the proposed filter.

VII. CONCLUSION

A fully integrated non-frequency-translating TRX front end for cellular BSs in 22-nm FD-SOI technology covering more than two octaves of BW, from 1.25 to 5.5 GHz, has been presented. The TX part features two linear variable gain feedback PPA stages and a high-power PA, which uses integrated 6.5-V LDMOS CG transistors to enable a 7.4-V PA supply. In addition, the TX benefits from using a 12.5- Ω differential impedance in providing a wideband frequency

response, a record high P_{sat} of 38.1 dBm, and a P_{avg} of 32.1 dBm for a 100-MHz 16-QAM OFDM 5G NR signal at 3.4 GHz with an EVM of -18 dB. Multiple design choices were made to thermally handle the high on-chip power dissipation, using reduced die thickness, exposed die package, liquid metal TIM, and pulsed measurements. A derivation of the PA hot-spot temperature increase has been made and confirmed by measurements using an on-chip temperature sensor. Furthermore, the TX is equipped with a filter for suppressing spurious signals, which realizes an RF notch using a cancellation path with a frequency-translated LPF. The filter can pass a transmitted 100-MHz 64-QAM OFDM 5G NR signal with low distortion and simultaneously suppress a large CW tone by >43 dB. The RX part features a differential LNA, matched to 12.5 Ω to use the same interface as the TX, a VGA, and a passive equalizer. Despite challenges at low impedance, the front-end NF is 1.3 dB thanks to the LNA and VGA architecture. A VGA gain reduction of 18 dB, from 29 to 10 dB, results in an NF increase of only 6 dB. Furthermore, the IIP3 outside the passbands is improved by up to 8 dB thanks to the proposed equalizer structure.

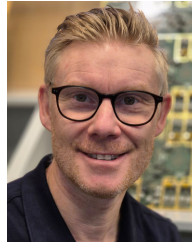
ACKNOWLEDGMENT

The authors would like to acknowledge Hans Hagberg for his strong support with pre-distortion of TX input signals.

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Florent Torres received the master's degree in electrical engineering and the Ph.D. degree in electronics from the University of Bordeaux, Bordeaux, France, in 2014 and 2018, respectively.

During his doctoral studies, he conducted research in collaboration with the IMS Laboratory, Bordeaux, and STMicroelectronics, Crolles, France, and focused on the design of power amplifiers for 5G applications in 28-nm fully depleted silicon on insulator (FD-SOI) technology. He is currently a Senior Researcher with the RF Frontend and PA Team, Integrated Radio System Research Group, Ericsson Research, Lund, Sweden, where he joined in 2018. His current research interests include the design of power amplifiers and front ends at RF and millimeter-wave frequencies using advanced technology nodes, device hardware research, and RF fingerprinting for physical layer security and trustworthiness applications.



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Martin Hansson received the M.Sc. degree in applied physics and electrical engineering and the Ph.D. degree in electrical components from Linköping University, Linköping, Sweden, in 2003 and 2008, respectively.

He participated in multiple industrial internships at Intel, Hillsboro, OR, USA, during his Ph.D. graduate studies. Since 2008, he has been with Ericsson, Lund, Sweden, working with HW product design and development, mainly within radio PCB and ASIC packages for radio base station, where he is currently a Senior Specialist, with a special interest in electrical package architecture and design as well as RF transmission design and simulation.



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Dr. Mattisson was a member of the Royal Swedish Academy of Engineering Sciences and the Royal Physiographic Society in Lund. He has been serving as the Technical Program Committee Member for the International Solid-State Circuits Conference and the European Solid-State Circuits Conference.



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Prof. Sjöland was a Technical Program Committee Member of European Solid-State Circuits Conference (ESSCIRC) from 2011 to 2016 and has been a Technical Program Committee Member of International Solid-State Circuits Conference (ISSCC) since 2024. He was an Associate Editor of IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS—II: EXPRESS BRIEFS from 2014 to 2015 and IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS—I: REGULAR PAPERS from 2016 to 2023 and is currently an Associate Editor of IEEE OPEN JOURNAL OF THE SOLID-STATE CIRCUITS SOCIETY.